

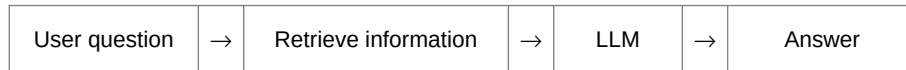
Introduction to LLM Pipelines

RAG (Retrieval-Augmented Generation) and LangSmith tracing

1. What is an LLM pipeline?

An LLM pipeline is a sequence of steps that takes an input from a user, processes that input, sends useful information to a language model, and returns a response. Instead of thinking of an AI application as only an LLM, it is often more useful to think of it as several connected components.

A simple pipeline might look like:



2. What is RAG?

RAG stands for Retrieval-Augmented Generation. In simple terms, RAG allows an LLM to look up relevant information from an external source before generating its answer.

A normal LLM mainly relies on the information it learned during training. A RAG system adds a retrieval step so the model can use information from a specific knowledge base, such as a PDF, database, website, or collection of documents.

3. How RAG works

- 1. User asks a question** — The user provides a question, such as: “What does tracing mean?”
- 2. Retrieve** — The system searches a knowledge base for chunks of text that are semantically relevant to the question.
- 3. Augment** — The retrieved text is added to the prompt as context for the LLM.
- 4. Generate** — The LLM uses the question and retrieved context to produce an answer.

Example

Suppose a PDF contains a section explaining LangSmith tracing. A user asks, “Why is tracing useful?” The RAG system searches the document, retrieves the section about tracing, places that text into the prompt, and asks the LLM to answer using the retrieved context.

4. Why use RAG?

- It allows an LLM application to use information that is specific to a document or knowledge base.
- The knowledge source can be updated without retraining the entire language model.
- It can reduce the chance that the model answers without relevant information.
- It is useful for document question-answering systems, internal knowledge assistants, and research tools.

5. Important RAG concepts

Document: The original source of information, such as a PDF or text file.

Chunk: A smaller section of a document. Documents are usually split into chunks so relevant sections can be retrieved.

Embedding: A numerical representation of text that captures semantic meaning. Similar pieces of text tend to have similar embeddings.

Vector database / knowledge base: A system that stores embeddings and allows the application to search for semantically relevant content.

Retrieval: The process of finding the most relevant chunks for a user's question.

Context: The retrieved information provided to the LLM so it can use the source material when generating its answer.

6. What is LangSmith?

LangSmith is a platform for observing, testing, and evaluating LLM applications. One of its key features is **tracing**, which records the steps involved when an AI application processes a request.

7. What does tracing mean?

Tracing means recording the sequence of operations performed by an AI pipeline. Instead of only seeing the final answer, a developer can inspect what happened inside the application.

For a RAG application, a trace might show the user's question, the retrieval step, the documents or chunks that were retrieved, the prompt sent to the model, the model's response, and timing or other execution information.

8. Why is tracing useful?

- **Debugging:** it helps identify which step caused a problem.
- **Retrieval inspection:** developers can check whether the RAG system retrieved the correct document chunks.
- **Prompt inspection:** developers can see what context was actually passed to the LLM.
- **Performance analysis:** traces can help identify slow or expensive steps in a pipeline.
- **Improvement:** by inspecting failed or poor responses, developers can change retrieval, prompts, models, or other parts of the pipeline.

9. Example: debugging a RAG chatbot with LangSmith

Imagine a user asks a question that is clearly answered in the PDF, but the chatbot gives an incorrect answer. The final response alone does not tell the developer why this happened.

A LangSmith trace can help break down the problem. If the correct PDF chunk was never retrieved, the problem may be with retrieval or embeddings. If the correct chunk was retrieved but the prompt did not include it correctly, the problem may be in the prompt or pipeline connections. If the correct context reached the LLM but the answer was still poor, the model or instructions may need improvement.

10. RAG and LangSmith together

RAG provides the mechanism for giving an LLM relevant external information, while LangSmith provides visibility into how the AI pipeline behaves. They solve different problems but work well together.

RAG	LangSmith
Retrieves relevant information	Records what happened in the pipeline
Provides context to the LLM	Shows the context and intermediate steps
Helps the model answer from a knowledge source	Helps developers debug and improve the system

11. Key takeaway

A RAG chatbot does not have to rely only on what the LLM already knows. It can retrieve relevant information from a knowledge base and give that information to the model as context. LangSmith tracing then allows developers to see how the pipeline handled the request, making it easier to find problems and improve the system.